

The Gradient of the Logistic Function

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1 Building Logistic Function

1.1 Data and Labels: d, x_j, y

Take one row of observations. Let there be d observations; where each x_j is the z-score for j th observation (the standard deviations from the mean),

$$x_1, x_2, \dots, x_d$$

and a single classification observation y where

$$y \in \{0, 1\}$$

You can think of y as yes or no, or true or false.

1.2 Linear Score: z, b

Our goal is simply take these x_j and predict y as best as possible. So intuitively we can take a single observation call it x_2 . Choose an arbitrary break $x_2 > 1$ say. Let our prediction by 1 for y when $x_2 > 1$. But this doesn't take into account all our observations. We want to take into consideration all of our variables when making a decision. So let z be the summation of the d observations.

$$z = x_1 + x_2 + \dots + x_d + b$$

We introduced a b which is a bias for the guess we want. You can imagine b as,

if all x_j are 0 then what would we like our guess be.

1.3 Input Weighting: w_j

Now it might be the case that some observations are more important than others, or interact with y in different ways. For example your length of credit history might be more important than your hair color in predicting if you will repay or not.

So assign a weight w_j to each observation x_j . So we have now

$$z = w_1x_1 + w_2x_2 + \dots + w_dx_d + b \quad (1)$$

But how do we get these weights? We need to conclude based on the current weights how confident we are that $y = 1$ ie given our current value of z how confident are we that this observation is a true $y = 1$. Well z is arbitrary, w_1 can be as big as we want, z can be 100, -0.5 so on; what does that mean?

1.4 Score to Probability: $\sigma(z)$

Take a function

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (2)$$

and let this be the predicted probability that $y = 1$. Specifically let

$$p = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (3)$$

To get a concept of a probability we must have the function $0 \leq p \leq 1$. We would also like $p \rightarrow 0$ as $z \rightarrow -\infty$ and $p \rightarrow 1$ as $z \rightarrow \infty$.

2 Properties

2.1 Bounds on p

Observe that since $e^x > 0$ always we have

$$1 + e^{-z} > 1 \iff \frac{1}{1 + e^{-z}} < 1$$

and

$$1 + e^{-z} > 0 \implies \frac{1}{1 + e^{-z}} > 0$$

Since denominator will always be greater than numerator and positive.

2.2 Limits of p

Observe that for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that whenever $z \geq N$ it follows

$$\left| \frac{1}{1 + e^{-z}} - 1 \right| < \epsilon$$

specifically choose

$$N > \log\left(\frac{1}{\epsilon}\right)$$

To see this

$$\begin{aligned} \left| \frac{1}{1 + e^{-z}} - 1 \right| &= \left| \frac{1}{1 + e^{-z}} - \frac{1 + e^{-z}}{1 + e^{-z}} \right| \\ &= \left| \frac{-e^{-z}}{1 + e^{-z}} \right| \\ &= \frac{e^{-z}}{1 + e^{-z}} \end{aligned}$$

Where the last line follows since $e^x > 0$ always.

Which also gives

$$1 + e^{-z} > 1$$

giving

$$\frac{e^{-z}}{1+e^{-z}} < e^{-z}$$

Now

$$\begin{aligned} e^{-z} < \epsilon &\iff -z < \log(\epsilon) \\ &\iff z > -\log(\epsilon) \\ &\iff z > \log\left(\frac{1}{\epsilon}\right) \end{aligned}$$

Meaning whenever $z \geq N$ where $N > \log\left(\frac{1}{\epsilon}\right)$ we have

$$\left| \frac{1}{1+e^{-z}} - 1 \right| < \epsilon$$

giving

$$p \rightarrow 1 \text{ as } z \rightarrow \infty$$

For the other direction observe that for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that whenever $z \leq -N$ it follows

$$\left| \frac{1}{1+e^{-z}} - 0 \right| < \epsilon$$

specifically choose $N > \log\left(\frac{1-\epsilon}{\epsilon}\right)$

To see this

$$\left| \frac{1}{1+e^{-z}} - 0 \right| = \frac{1}{1+e^{-z}}$$

Which follows from $e^x > 0$ always. Again giving

$$1 < \epsilon(1+e^{-z})$$

and

$$\frac{1-\epsilon}{\epsilon} < e^{-z}$$

taking log of both sides and multiplying by -1

$$\log\left(\frac{1-\epsilon}{\epsilon}\right) < -z \iff -\log\left(\frac{1-\epsilon}{\epsilon}\right) > z$$

Meaning whenever $z \leq -N$ where $N > \log\left(\frac{1-\epsilon}{\epsilon}\right)$ we have

$$\left| \frac{1}{1+e^{-z}} - 0 \right| < \epsilon$$

giving

$$p \rightarrow 0 \text{ as } z \rightarrow -\infty$$

Now that we have proved the limit behavior of p and we have defined it as the probability that $y = 1$ for our model, we can define the loss that we are looking to minimize.

2.3 Loss: L

Intuitively we would want the distance from our predicted probability to the true value of $y = 0$ or $y = 1$. But an absolute difference doesn't account for the confidence difference between a prediction of .99 and .999 for a $y = 1$, which are 0.01 and 0.001 away from 1 but are dramatically different in confidence. Therefore we define the loss using logs. Specifically we let the loss be

$$L = -\log(p)$$

if $y = 1$ and

$$L = -\log(1-p)$$

if $y = 0$.

In closed form our loss looks like

$$L = -[y \log(p) + (1 - y) \log(1 - p)] \quad (4)$$

3 Optimization

3.1 Gradient: $\frac{\partial L}{\partial w_j}$

As stated before we want to minimize our loss. So how can we do this. We want to tune our weights as optimally as possible. Therefore we use the gradient specifically

$$\frac{\partial L}{\partial w_j}$$

But we defined L in terms of y and p so we must chain through to get our derivative.

Observe by the chain rule

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial p} \frac{\partial p}{\partial z} \frac{\partial z}{\partial w_j}$$

Moving from right to left we have

$$\frac{\partial z}{\partial w_j} = x_j$$

since taking the partial derivative leaves all other weights as constants

Next

$$\begin{aligned} \frac{\partial p}{\partial z} &= \sigma'(z) \\ &= -1(1 + e^{-z})^{-2}(-e^{-z}) \\ &= \frac{e^{-z}}{(1 + e^{-z})^2} \end{aligned}$$

Observe that

$$\frac{e^{-z}}{(1 + e^{-z})^2} = \frac{e^{-z}}{(1 + e^{-z})} \frac{1}{(1 + e^{-z})}$$

where the right side is again $\sigma(z)$ and the left side is $1 - \sigma(z)$ since

$$\begin{aligned} 1 - \sigma(z) &= 1 - \frac{1}{1 + e^{-z}} \\ &= \frac{1 + e^{-z}}{1 + e^{-z}} - \frac{1}{1 + e^{-z}} \\ &= \frac{e^{-z}}{(1 + e^{-z})} \end{aligned}$$

Giving

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

Finally

$$\frac{\partial L}{\partial p} = \frac{-y}{p} + \frac{1 - y}{1 - p}$$

Now chaining all three together gives

$$\begin{aligned} \left(\frac{-y}{p} + \frac{1 - y}{1 - p} \right) p(1 - p)x_j &= (-y(1 - p) + (1 - y)p)x_j \\ &= (-y + py + p - py)x_j \\ &= (p - y)x_j \end{aligned}$$

giving the gradient as

$$(p - y)x_j \tag{5}$$

For b simply observe

$$\frac{\partial z}{\partial b} = 1$$

and therefore the gradient is

$$(p - y) \tag{6}$$

We now have the optimal unit of movement of each w_j in order to most efficiently decrease our loss.

3.2 Averaging over a Dataset

In general we have d columns of observations and n rows of data. We can then simply think of each variable described above as a vector. Where w is the weight vector in which we add b . p is the vector of probabilities one for each row p_i . y is the vector of observations of y_i . To get our final gradient we take loss as the average loss of the rows.

$$L = \frac{1}{n} \sum_{i=1}^n L_i \tag{7}$$

where

$$L_i = -[y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Then to find the partial derivative of the gradient we have

$$\frac{\partial L}{\partial w_j} = \frac{1}{n} \sum_{i=1}^n \frac{\partial L_i}{\partial w_j}$$

since the partial derivative of the mean is the mean of the partial derivatives. Our model reaches its optimum specifically when

$$\frac{\partial L}{\partial w_j} = 0 \tag{8}$$

and

$$\frac{\partial L}{\partial b} = 0$$

for all weights w_j and bias b .

3.3 Optimum

In general at optimum we have

$$\sum_{i=1}^n p_i = \sum_{i=1}^n y_i \tag{9}$$

Which follows from

$$\begin{aligned} \frac{\partial L}{\partial b} = 0 &\iff \frac{1}{n} \sum_{i=1}^n \frac{\partial L_i}{\partial b} = 0 \\ &\iff \frac{1}{n} \sum_{i=1}^n (p_i - y_i) = 0 \\ &\iff \frac{1}{n} \sum_{i=1}^n p_i - \frac{1}{n} \sum_{i=1}^n y_i = 0 \\ &\iff \frac{1}{n} \sum_{i=1}^n p_i = \frac{1}{n} \sum_{i=1}^n y_i \\ &\iff \sum_{i=1}^n p_i = \sum_{i=1}^n y_i \end{aligned}$$

Where our logistic regression will produce its most optimal predictions based on our assumptions and data input.

We can then use the model to predict further classifications by training on data we believe to be representative of the further data we are trying to predict.